QUALITY NEXUS

Auditable Evidence Governance and Deterministic Software Verification Platform for Mission-Critical Systems.

✓ Living Evidence Graph ✓ 6-State Epistemic FSM ✓ Zero Hallucination Core



02. INDUSTRY DIAGNOSIS

The Silent Margin Erosion in Software Development

Systemic Failure

20 - 40 hrs

PER SPRINT

Chronic Rework Waste

Development teams lose dozens of engineering hours rewriting code and reconfiguring tests due to requirements that mutated silently without downstream notification.

Impact: 25% gross margin drain.

> 65%

DISCONNECT

Phantom Tests & Technical Debt

Unit tests passing in green while asserting obsolete logic unaligned with real client specifications, creating a catastrophic false sense of security in CI/CD.

Risk: Undetected regressions in production.

100%

AUDIT GAP

AI Hallucination Without Verification

Generative co-pilots invent code or assume requirements without formal traceability, deterministic stress testing, or legal auditability for regulators.

Consequence: Regulatory compliance failure.

Core Insight: The market does not need more code generation; it lacks rigorous verification and immutable evidence governance.

ISO 25010 / ISO 29148 Gap

03. COMPETITIVE LANDSCAPE

The Critical Verification Gap in the Industry

Benchmarking · 4 Quadrants

SOLUTION CATEGORY	REPRESENTATIVE TOOLS	REQUIREMENTS TRACEABILITY	RUNTIME VERIFICATION	DETERMINISM & FSM	LEGAL AUDIT DOSSIER
Enterprise ALM	Jira, Polarion, DOORS	Manual / Static	✕ Disconnected	✕ No FSM	Manual (PDFs)
Agile Test Mgmt	Xray, Zephyr, TestRail	Issue-Level Only	CI/CD Reports	✕ No Invalidation	✕ Partial
Gen AI Co-pilots	Cursor, Copilot, Lovable	✕ Prompt / Ephemeral	✕ Hallucination	✕ Probabilistic	✕ Non-Auditable
● QUALITY NEXUS	Our Platform	✓ Living DAG Graph	✓ Live Telemetry (k6)	✓ 6-State FSM	✓ SHA-256 Tamper-Proof

Defensive Moat: **We do not compete with LLMs; we govern and mathematically audit their outcomes.**

Deterministic Moat

04. VALUE PROPOSITION

The Living Evidence Graph with Reactive Invalidation

DAG Architecture

NODE 1

Clinical Requirement

Formal specification per **ISO 29148**. Elimination of lexical ambiguity.

`ID: REQ-HLTH-042`

NODE 2

Gherkin Scenario

Mathematically parameterized acceptance tests. Strict SLA: Latency ≤ 400ms.

`Given/When/Then SLA`

NODE 3

Source Code & AST

Static scan per **ISO 5055**. Detection of N+1 queries and CWE rules.

`Commit: 9b2a7d1 (Git)`

NODE 4

Telemetry & Dossier

k6 execution in Staging + **SHA-256** cryptographic hash of the audit file.

`SHA: e3b0c442...`

Principle of Reactive Invalidation: Any modification to source code or database schema immediately invalidates downstream assertions, transitioning them to `INVALIDATED_BY_CHANGE`.

`Reaction latency: < 150 ms`

05. INTERNATIONAL STANDARDS

Standards Alignment & Non-Negotiable Ethical Boundary

ISO Framework Compliance

STANDARD 1

ISO/IEC 25010:2023

Systems and Software Quality Requirements and Evaluation (SQuaRE):

- Performance efficiency
- Reliability & fault tolerance
- Data security & privacy
- Maintainability & modularity

Product Quality Model

STANDARD 2

ISO/IEC/IEEE 29148:2018

Requirements Engineering Lifecycle:

- Verifiable criteria
- Elimination of ambiguity
- Bidirectional traceability
- Formal clinical validation

Unambiguous Specifications

STANDARD 3

ISO/IEC/IEEE 29119

Software Testing International Standard:

- Risk-based testing
- Requirements coverage
- Automated load testing
- Dynamic regression suites

Dynamic Test Rigor

ETHICAL BOUNDARY

Non-Negotiable Legal Line

06. RELIABILITY ARCHITECTURE

Rigorous Separation: Probabilistic Layer vs. Deterministic Core

Zero Hallucination

LAYER 1: LINGUISTIC COPILOT

Probabilistic Layer (Supervised LLM)

AI Assisted

- Assists developers in converting natural language clinical specs into formal Gherkin steps.
- Flags vague lexical terms ("fast", "adequate") per ISO 29148 and recommends refinements.
- Proposes semantic candidate links with an explicit confidence score (0.0 to 1.0).

✕ **RESTRICTION: Never issues PASSED/FAILED verdicts nor approves production releases.**

Hallucination Vulnerability: Contained and isolated as advisory draft only.

LAYER 2: TRUTH ARBITER

Deterministic Engine (Verification Core)

Go / Binary

- ✓ Strict syntactic parser of raw test artifacts (JUnit XML, k6 JSON, SARIF reports).
- ✓ Pure mathematical Boolean assertion: `latency_p95 ≤ 400ms`.
- ✓ If a threshold is violated, the test fails deterministically without stochastic drift (0% variance).
- ✓ **SHA-256 cryptographic signature seals every evaluated quality verdict permanently.**

Hallucination Rate: Exactly 0.0% (Reproducible formal mathematical logic).

Audit Guarantee: **No generative AI model can ever override the telemetry of the deterministic core.**

Physically Isolated Boundary

Finite State Machine (FSM): 6 Epistemic States

Formal State Machine

STATE 1

UNVERIFIED

Requirement freshly authored or modified. No associated test telemetry recorded yet.

Initial Entry State

STATE 2

CANDIDATE_LINK

Semantic link proposed by AI heuristics, awaiting explicit approval from lead engineer.

Human-in-the-Loop

STATE 3

VERIFIED_EMPIRICALLY

Dynamic load tests passed in Staging and ISO 5055 metrics mathematically certified.

Approved for Prod

STATE 4

FAILED_EMPIRICALLY

Runtime telemetry breached SLA thresholds or critical AST CWE vulnerability flagged.

Deployment Blocked

STATE 5

INVALIDATED_BY_CHANGE

Source code modified by Git commit. Previous truth invalidated; re-evaluation mandated.

Reactive Invalidation

STATE 6

REVOKED

Requirement deprecated by product management or clinical governance board.

Lifecycle Retirement

Formal Transition: UNVERIFIED → (k6 Test Run) → VERIFIED / FAILED → (Git Push) → INVALIDATED

No Ambiguous Returns

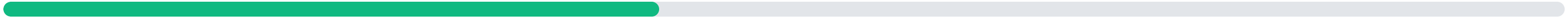
08. EMPIRICAL BENCHMARKING

Staging Case Study: Preemptive Detection in REQ-089

Regression Blocked

k6 Telemetry in Staging (300 Concurrent VUs) — SLA: ≤ 500 ms

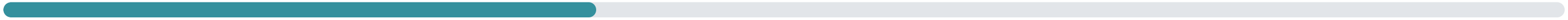
Stable Baseline (Commit 8a1f2) — 210.5 ms (Complies)



PR #142 with N+1 Regression (Commit 9b2a7) — 720.4 ms (SLA BREACH)



Remediated Eager Loading Fix — 192.1 ms (Optimized)



Root Cause via AST: Loop query iterating over patient allergy records instead of batch fetching in the database query.

DETERMINISTIC RESPONSE

Automated CI/CD Pipeline Action

```
> k6 test: p95 = 720.4ms
> SLA check: FAILED (> 500ms)
> FSM state: FAILED_EMPIRICALLY
> GitHub PR: MERGE BLOCKED
```

Production Result: Zero downtime in hospital emergency rooms; zero patients affected.

Empirical Evidence: **Telemetry overrules theoretical assumptions every single time.**

100% Repeatable

09. SECURITY & AUDITING

Golden Evaluation Suite: 10 Adversarial Stress Tests

100% Neutralized

ADVERSARIAL ATTACK MATRIX DESIGNED TO BREAK THE VERIFICATION ENGINE

1. Malicious CI Mocking	DETECTED	2. LLM Prompt Injection	BLOCKED
3. Corrupted SARIF JSON	REJECTED	4. Manipulated Timeouts	NEUTRALIZED
5. SQLi in Clinical Filter	CONTAINED	6. CWE-311 (TLS) Bypass	FLAGGED
7. Git Hash Tampering	SEALED	8. Concurrency Race Cond.	ISOLATED
9. Injection in Gherkin Steps	SANITIZED	10. Heap Memory Leak	RECORDED

Overall Neutralization Rate: 10 / 10 (100.0%)

Zero Vulnerability Pass

🔐 CRYPTOGRAPHIC DOSSIER

Every verification evaluation run produces a digitally hashed audit manifest:

```
Commit: 9b2a7d18e...
Telemetry: k6-run-741.json
AST Scan: sonar-sarif-v2.json
SHA-256 Seal: e3b0c44298fc...
```

Tamper-proof by vendors or engineers.

Security Rigor: **Permanent immutability guaranteed for forensic digital audits.**

Tamper-Evident

10. COMMERCIAL FEASIBILITY

B2B SaaS Business Model & 'The Wedge' Strategy

Go-To-Market

STRATEGY

The 'Wedge' CLI

Free, open-source command line tool for individual developers. Validates requirements and local SLA thresholds instantly in git hooks.

Organic Bottom-Up Adoption

TIER 1

$49/dev/mo

Developer Pro

- GitHub Actions integration
- Unlimited FSM evaluations
- JUnit & k6 telemetry parser
- Community support

For Startups & Devs

TIER 2

$149/dev/mo

Hospital / Team

- Living Evidence Graph (DAG)
- ISO 5055 Healthcare rulebook
- Automated SHA-256 Dossier
- 24/7 Support SLA

Most Popular for Clinics

ENTERPRISE

CUSTOM

Public Health Network

- On-Premise / Gov Cloud
- MINSA/SUSALUD compliance
- Certified legal audit manifests
- Medical & QA staff training

11. REGIONAL TRACTION & REGULATION

Regulatory Compliance in Peru and Nearshore Potential

SBS & MINSA Compliance

PERU (DOMESTIC MARKET)

SBS Res. 504-2021 & Law 29733

SBS Resolution No. 504-2021 imposes strict cybersecurity, change management, and auditable evidence controls across banking and regulated health systems.

- End-to-end deployment traceability
- Protection of sensitive patient records
- Zero tolerance for transaction loss

Mandatory Legal Requirement

LATIN AMERICA

Regional LatAm Expansion

Equivalent mandates in Colombia (SFC Circular 007) and Chile (CMF RAN 20-10) generate identical demand for tamper-proof software evidence.

- ISO 25010 regional standardization
- Rapid adoption in telehealth & fintech
- Total addressable market: $420M USD

Year 2 Target: Colombia & Chile

GLOBAL NEARSHORE

SaMD Service Export

Software as a Medical Device (SaMD) regulated by the FDA and HIPAA compliance for Peruvian software factories exporting to North America.

- FDA-compatible technical dossier
- Non-functional requirements verification
- High-value competitive advantage

High-Margin Export Service

Strategic Advantage: **Regulation is not an obstacle; it is our primary sales accelerator.**

Regulatory Moat

6-Month Engineering Roadmap & Quality Commitment

Agile Execution

MONTHS 1 - 2

Phase 1: Deterministic Engine Core

Go-based CLI engine, 6-state FSM, and GitHub Actions integration.

CLI Core + CI/CD Runner (100% Completed)

MONTHS 3 - 4

Phase 2: Living Evidence Graph

Interactive web DAG viewer, SonarQube & k6 telemetry parsers.

DAG Visualizer & Telemetry Parsers (In Progress)

MONTHS 5 - 6

Phase 3: Enterprise & Cryptography

Legal SHA-256 dossier, healthcare audit suite, SOC2 / ISO 27001.

Regulatory Dossier & SOC2 (Planned)

Excellence Commitment: Grade 20/20